THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 15

Unity Catalog vs Hive Metastore — The Migration Interview Question

SECTION 01

Why This Comparison Comes Up So Often

Most organizations running Databricks today didn't start with Unity Catalog. They started with the built-in, workspace-level Hive metastore, and are now somewhere in the process of migrating. This makes the comparison a natural, practical interview topic, especially for roles touching platform or governance work.

CORE CONCEPT

Although Databricks workspaces continue to include the built-in Hive metastore, data governance using Hive metastore is deprecated, and Databricks recommends migrating all data from the legacy Hive metastore to Unity Catalog.

SECTION 02

Core Differences

Aspect	Hive Metastore	Unity Catalog
Scope	Workspace-level object	Account-level object
Namespace	Two-level: database.table	Three-level: catalog.schema.table
Cross-Workspace Sharing	Not natively supported; each workspace isolated	Native; one metastore shared across many workspaces
Governed Asset Types	Tables and views only	Tables, views, ML models, dashboards, files, functions, volumes
Lineage and Auditing	Not natively provided	Built-in, unified lineage and audit logging
Access Control Groups	Local, workspace-specific users and groups	Account-level groups, consistent across workspaces

SECTION 03

How Access Control Actually Differs

IMPORTANT NUANCE

The Unity Catalog is an account-level object, and the Hive metastore is a workspace-level object. Permissions defined within the hive_metastore catalog always refer to the local users and groups in the workspace, while access control policies in Unity Catalog are applied to account groups instead of workspace-local groups.

This means a permission granted in one workspace's Hive metastore has no bearing on any other workspace, since the underlying users and groups are locally scoped. Unity Catalog's account-level groups solve this by giving the same identity a consistent set of permissions no matter which linked workspace they're working from.

SECTION 04

What Migration Actually Involves

- No hard cutover required: Unity Catalog is additive and can coexist with the Hive metastore, which continues to be accessible as the hive_metastore catalog.
- Partition handling changes: Unity Catalog manages partitions differently than Hive, and Hive commands that directly manipulate partitions are not supported on tables managed by Unity Catalog.
- Managed tables always use Delta: when upgrading a Hive table to a Unity Catalog managed table, the result always uses the Delta table format, and Unity Catalog fully manages its lifecycle, file layout, and storage.
- Table history is not preserved automatically: table history is not migrated when you run CREATE TABLE CLONE as part of an upgrade.
- Federation as an alternative: Hive metastore federation lets you create a Unity Catalog catalog that mirrors your existing Hive metastore, without a full table-by-table migration.

SECTION 05

When Hive Metastore Might Still Be Used

- Running a single, legacy Hadoop or Spark environment with no near-term plans to migrate.
- Very simple data needs, with no requirement for advanced security, sharing, or governance across teams.
- Early stages of a Databricks adoption, using the built-in Hive metastore as a temporary step before a planned upgrade to Unity Catalog.

GOOD TO KNOW

By default, compute clusters continue to connect to the Hive metastore even after Unity Catalog is adopted, unless access is explicitly disabled workspace-wide or per cluster.

SECTION 06

Scope Comparison at a Glance

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HIVE METASTORE (workspace-scoped, isolated)
Workspace A | Workspace B | Workspace C
own metastore, own local users/groups each
(no native sharing between workspaces)

UNITY CATALOG (account-scoped, shared)
ONE METASTORE + account-level groups & permissions
      |            |            |
 Workspace A   Workspace B   Workspace C
(shared view, shared governance, shared lineage)
```

REMEMBER THIS FOR INTERVIEWS

The single biggest architectural difference is scope: Hive metastore is workspace-level and isolated, Unity Catalog is account-level and shared. Every other difference, namespace depth, governed asset types, and group model, follows from that core distinction.

SECTION 07

Glossary — Day 15 Terms

Term	Definition
Hive Metastore	The legacy, workspace-level metadata store built for Hadoop-era workflows; now deprecated for governance.
hive_metastore Catalog	The special catalog name under which the legacy Hive metastore is surfaced inside Unity Catalog's namespace.
Account-Level Group	A group defined at the Databricks account level, used consistently for access control across all linked workspaces in Unity Catalog.
Hive Metastore Federation	A migration approach that creates a Unity Catalog catalog mirroring an existing Hive metastore, without a full table-by-table migration.
Partition Manipulation	Direct commands to add, drop, or alter partitions; supported in Hive but not on Unity Catalog managed tables.

SECTION 08

Interview Questions & Answers

Q1. What is the core scope difference between Hive Metastore and Unity Catalog?

A. Hive Metastore is a workspace-level object, isolated per workspace, while Unity Catalog is an account-level object that can be shared across every workspace in a region through a single metastore.

Q2. Why can't permissions granted in one workspace's Hive metastore apply to another workspace?

A. Because permissions defined within the hive_metastore catalog always refer to local users and groups scoped to that specific workspace, whereas Unity Catalog uses account-level groups that carry the same permissions across all linked workspaces.

Q3. Does adopting Unity Catalog require immediately abandoning the Hive metastore?

A. No. Unity Catalog is additive and can coexist with the Hive metastore, which remains accessible as the hive_metastore catalog, and by default clusters continue to connect to it unless access is explicitly disabled.

Q4. What data asset types does Hive Metastore govern compared to Unity Catalog?

A. Hive Metastore covers tables and views only, while Unity Catalog extends governance to ML models, AI assets, dashboards, and files as well.

Q5. What changes when a Hive table is upgraded to a Unity Catalog managed table?

A. The resulting managed table always uses the Delta table format, and Unity Catalog fully manages its lifecycle, file layout, and storage; however, table history is not migrated when using CREATE TABLE CLONE as part of the upgrade.

Q6. What is Hive metastore federation?

A. An alternative migration approach that creates a catalog in Unity Catalog mirroring an existing Hive metastore, allowing Unity Catalog to govern the tables registered there without requiring a full table-by-table migration.

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